DFT-D3 for Solid-State Materials

A Data-Driven Perspective on Accuracy

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Why Accurate Interaction Calculations Are Important:

The paper starts by saying that accurately predicting atomic interactions is essential for:

- understanding chemical bonding
- predicting material properties
- designing new materials

One of the most important weak interactions is van der Waals (vdW) dispersion interactions.

These interactions are important in:

- molecular crystals
- layered materials (like graphite)
- adsorption on surfaces

Dispersion interactions arise due to instantaneous dipole fluctuations between atoms.

They decay with distance as: $E \propto R^{-6}$

R = interatomic distance.

Dispersion Interactions:

Dispersion interactions originate from electron correlation effects.

High-accuracy quantum chemistry methods such as Coupled Cluster (CC) can capture dispersion naturally. However,

$$\text{Computational cost} \propto N^7$$

This makes them impractical for large molecules or periodic solids.

So, to counter large systems we use **DFT(Density Functional Theory)**

DFT is popular because:

- good balance between accuracy and speed
- works for large systems

Limitations in DFT:

Common exchange–correlation functionals like, **GGA** (Generalized Gradient Approximation) and **hybrid functionals** can not capture dispersion interactions properly.

This leads to errors in:

- binding energies
- lattice constants
- cohesive energies.

To fix this limitation, researchers add an extra dispersion correction term.
Total energy becomes:

$$E_{\text{Total}} = E_{\text{Base}} + E_{\text{disp}}$$

E base = energy from the functional
E disp = correction for dispersion.

Various **DISPERSION MODELS:**

Various schemes have been developed to compute E disp, broadly categorized into pairwise additive and many-body approaches.

Most widely used pairwise dispersion correction schemes is Grimme's DFT-D series, which comprises variants **D2, D3,** and **D3 (BJ).**

In all DFT-D methods, the dispersion correction added to the self-consistent DFT energy is generally written as:

$$E_{disp} = -\sum_{n=6,8,10,\ldots} \sum_{A<B} s_n \frac{C_n^{AB}}{R_{AB}^n} f_{\mathrm{damp,n}}(R_{AB})$$

- R(ab) is the interatomic distance between atoms A and B
- CABn are the dispersion coefficients
- Sn are global scaling factors depending on the chosen DFA

1)DFT -D2

The expression for D2-dispersion correction is given as:

$$E_{\text{disp}}^{\text{D2}} = -s_6 \sum_{A<B} \frac{C_6^{AB}}{R_{AB}^6} f_{\text{damp}}(R_{AB})$$

with, $C_6^{AB} = \sqrt{C_6^A C_6^B}, \qquad C_6^a = 0.05\, N_p^a\, I_p^a\, \alpha^a$

- Na p is the effective number of electrons
- Iap is the ionization potential, and
- α a is the static polarizability of atom a

2)DFT -D3

The expression for D3 has Two body terms E(2) and Three body terms E(3) given as:

$$E_{disp}^{D3} = E^{(2)} + E^{(3)}$$

- Two-body term E(2) is expressed as:

$$E^{(2)} = -\sum_{n=6,8}\sum_{A<B} s_n \frac{C_n^{AB}(\mathrm{CN}^A, \mathrm{CN}^B)}{R_{AB}^n} f_{\mathrm{damp,n}}(R_{AB})$$

where each Cn is interpolated over reference values via the atoms coordination numbers (CN). Here, s6 is usually set to 1, whereas s8 is an adjustable parameter that is fitted for each density functional.

For the damping function,the widely used variant is the **zero- damping variant** of the D3 correction, **D3(0)** is given by:

$$f_{\mathrm{damp},n}^{\mathrm{D3(0)}}(r_{ij}) = \frac{1}{1 + 6\left(\frac{r_{ij}}{s_{R,n} R_{0ij}}\right)^{-\alpha_n}}$$

parameters $\alpha 6$ = 14, $\alpha 8$ = 16, and sR,8 = 1 are fixed, while sR,6 serves as a functional-dependent adjustable parameter.

2)DFT -D3

For the damping function,alternative and more widely adopted damping scheme is the **Becke–Johnson** damping function **D3(BJ)** is given by:

$$f_{\mathrm{damp},n}^{\mathrm{D3(BJ)}}(R_{AB}) = \frac{R_{AB}^{n}}{R_{AB}^{n} + (a_1 R_{0AB} + a_2)^n}$$

a1 and a2 are empirical damping parameters optimized against high-level reference interaction energies.

In the D3 framework,the dispersion coefficients C6AB, depend on the local chemical environment of atoms A and B.

The coordination number (CN) for atom A is defined as:

$$\mathrm{CN}^A = \sum_{B \neq A}^{N} \frac{1}{1 + \exp\left[-k_1\left(\frac{k_2(R_{A,\mathrm{cov}} + R_{B,\mathrm{cov}})}{r_{AB}} - 1\right)\right]}$$

- N is the total number of atoms, RA,cov and RB,cov are scaled covalent radii of atoms A and B
- the constants k1 = 16 andk2 = 4/3 are predefined.

2)DFT -D3

CAB 6 coefficient is then obtained via interpolation over reference systems as:

$$C_6^{AB}(\mathrm{CN}^A, \mathrm{CN}^B) = \frac{\sum_i^A \sum_j^B C_{\mathrm{ref}}^{AB}(\mathrm{CN}_i^A, \mathrm{CN}_j^B) L_{ij}}{\sum_i^A \sum_j^B L_{ij}}$$

C6 AB ref denotes precomputed reference dispersion coefficients for coordination environments **CNAi** and **CNBj**

the weighting factors **Lij** are defined by:

$$L_{ij} = \exp\left[-k_3\left((\mathrm{CN}^A - \mathrm{CN}_i^A)^2 + (\mathrm{CN}^B - \mathrm{CN}_j^B)^2\right)\right]$$

k3 = 4 controlling the sharpness of the interpolation.

3)Tkatchenko–Scheffler (TS):

Computes dispersion coefficients directly from the electron density. It employs a Hirshfeld-type partitioning to derive environment-dependent atomic polarizabilities:

$$\alpha_A = \alpha_A^{\text{free}} \left(\frac{V_A}{V_A^{\text{free}}} \right)$$

αA and VA represent the atomic polarizability and volume in the molecular environment, respectively

this enables the TS method to naturally capture effects such as oxidation states and charge transfer, making it more physically grounded for diverse chemical environments

Disadvantages of DFT-D methods:

- while pairwise dispersion correction schemes offer a favorable balance between computational efficiency and accuracy for a wide range of systems, but they inherently **neglect many-body polarization** and **screening effects**

  these omissions are crictical in extended solids and densely packed phases

- The **Many-Body Dispersion** (MBD) method addresses these limitations by representing atoms as **quantum harmonic oscillators** (QHOs) that interact via **long-range dipole–dipole** couplings.

$$H_{\mathrm{MBD}} = \sum_i \left(\frac{p_i^2}{2m} + \frac{1}{2} m\omega_i^2 x_i^2 \right) + \sum_{i<j} V_{\mathrm{dip}}(x_i, x_j)$$

Comparison b/w DFT-D and MBD methods:

- This **MBD method** formalism captures collective dipolar fluctuations and the resulting long-range correlation energy, offering improved accuracy for systems such as molecular crystals and van der Waals heterostructures.
- However, the increased computational cost associated with **MBD** renders it impractical for high-throughput screening and simulations of large-scale systems.
- In contrast, the **DFT-D3 method** correction scheme remains widely adopted due to its simplicity and computational efficiency.

Purpose of this Paper:

The main aim of this paper is to:

1. Study when D3 fails in solids
2. Identify origin of artificial minima
3. Compare D3 with many-body dispersion methods
4. Develop diagnostic tools to detect errors.

Computational Methodologies used:

- Electronic structure calculations were done with VASP using PAW method and PBE functional.
- Structural relaxations were performed with a plane-wave energy cutoff of 500 eV.
- Brillouin zone sampling was done using a Monkhorst–Pack k-point mesh with a reciprocal space resolution of 0.05 Å-1 in each direction.
- The study compares:D3 (ZD), D3 (BJ) and MBD as the reference method.
- distances scanned from 0.75 × Re to 2.5 × Re in increments of (0.05 × Re).

Dataset used:

- The dataset includes 400 Cu-, 321 K-, 453 Li-, 422 Mg-, 319 Mn-, and 165 Zn-containing structures.
- For each structure, dispersion energy contributions, interatomic distances, and coordination numbers were computed.

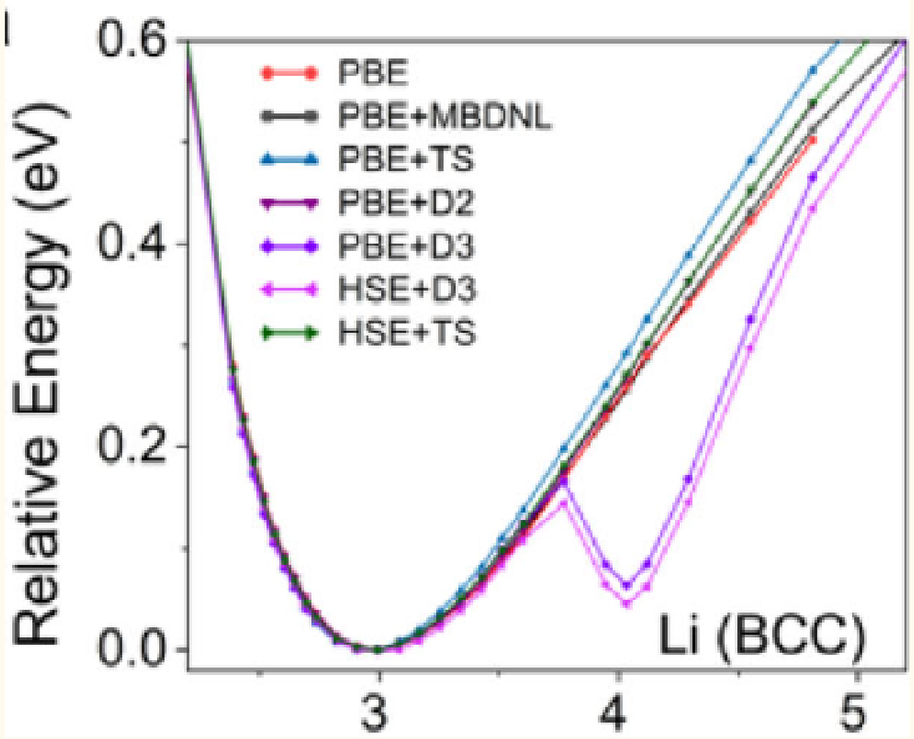
- Also collected atomic descriptors like: polarizability, ionic radius, electronegativity, ionization energy, electron affinity.
- These descriptors were used for correlation analysis.

Results and Discussion:

The authors studied how the the total energy of solids changes as the bond length is varied. this is known as analyzing the Potential Energy Surface (PES).

A PES represents:Energy of system vs atomic configuration

- In a physically correct system, this curve should be having one minimum, representing the equilibrium structure.
- but when DFT-D3 corrections are used, some systems show two minima instead of one and these appeared at larger bond distances.

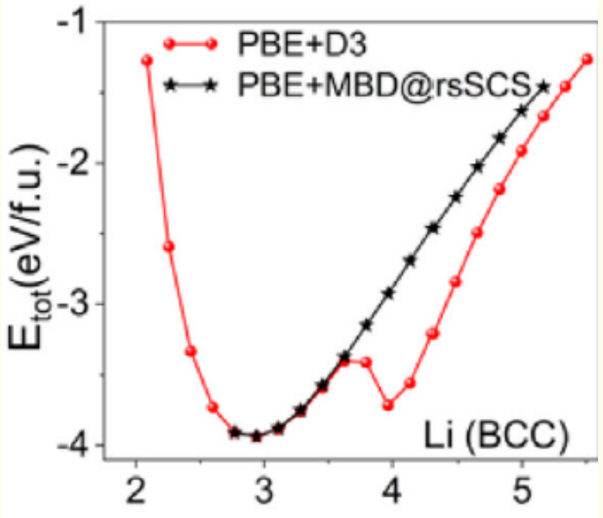


the second minima is visible while using d3 calculations, which shouldnt be there

Results and Discussion:

If you see in detail in these materials **DFT-D3** produce an extra energywell around ~4 Å. However, this minimum does not exist in reality.

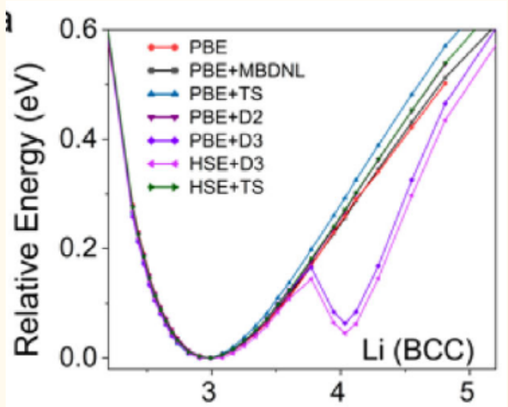
To verify whether the minimum is real, the authors compared **D3** with Many-Body Dispersion **(MBD)** calculations.



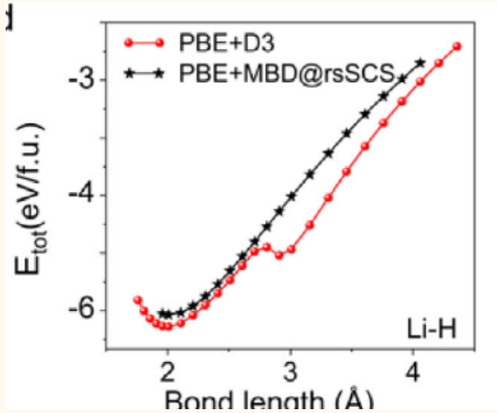
PBE+D3 shows **Two minima**, while PBE+MBD shows only **One minimum**. This clearly shows that the second minimum is an artifact introduced by D3

This second minimum is called Artificial or spurious minimum

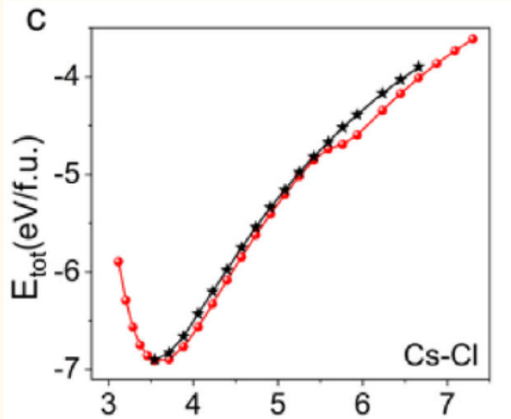
Examples of many systems:



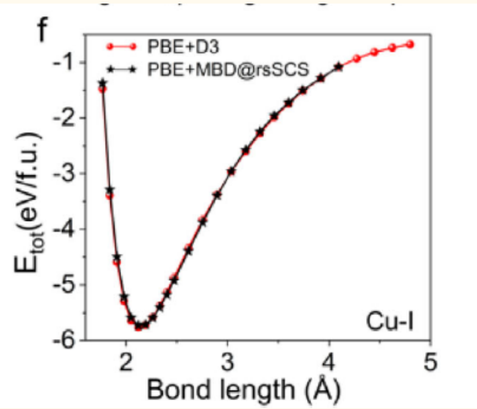
Lithium-BCC



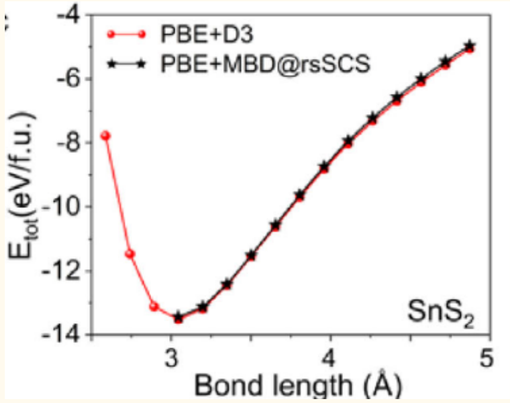
Li-H



Cs-Cl



Cu-I



SnS2

Why This Is a Serious Problem??

If a computational method predicts fake energy minima, it can lead to incorrect conclusions.

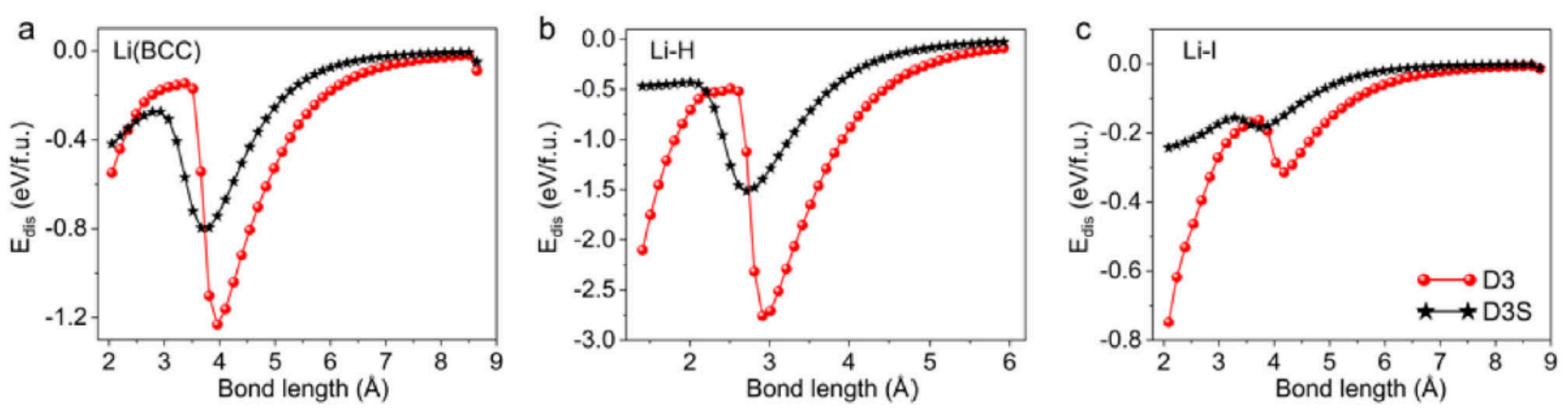
Possible consequences:

- Wrong lattice parameters
- Incorrect phase stability
- Artificial structural distortions
- Misleading thermodynamic predictions

For example:
A simulation might predict a structure that does not actually exist.

Evidence From Dispersion Energy:

The authors also analyzed the dispersion energy component alone.

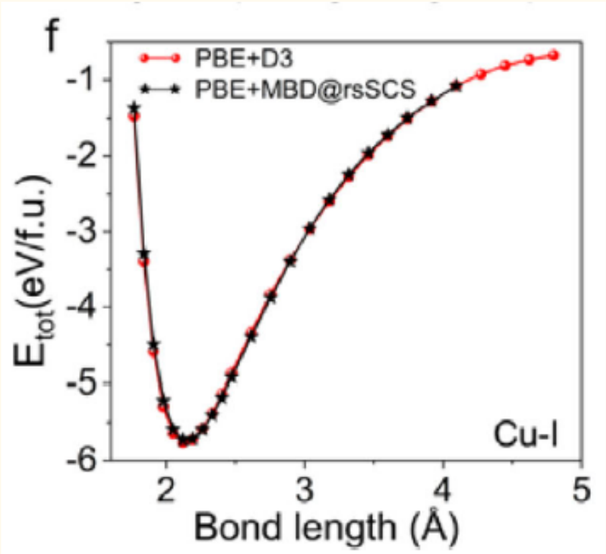


Comparison of dispersion energy (Edis) profiles obtained from DFT-D3S and DFT-D3 methods for range of solid systems.

For systems like Li-H, they observed that the dispersion term itself shows a minimum. This confirms that the artifact originates from the dispersion correction not from the underlying DFT functional.

System-Dependent Behavior:

Interestingly, the artifact does not appear in all systems.



The energy curve for CuI does not show a secondary minimum. This indicates that the problem is system-dependent

And at the same time when you compare it with MBD, the artifact disappears, therefore Artificial minima originate from the D3 dispersion term.

The authors reference earlier work by Tkachenko and Head-Gordon, which also reported similar spurious minima in dispersion-corrected DFT calculations. These studies suggest that the issue arises from limitations in pairwise dispersion models.

Key takeaways:

The key takeaway from this section is, DFT-D3 **can introduce artificial stabilization** in certain solids, leading to **unphysical minima** in the potential energy surface.

This effect is most pronounced in:

- metallic systems
- ionic systems
- highly polarizable atoms.

As we have observed these artificial minima, the next important question is what is the origin of this artifact in the D3 model?

Origin of the artifact:

Unlike earlier dispersion models, D3 does not assign fixed C6 values to atoms and the dispersion coefficient depends on the coordination number (CN) of atoms A and B. This means the dispersion strength changes depending on the local chemical environment.

Atoms in different environments have different:

- polarizabilities
- electronic screening
- bonding characteristics

Therefore the dispersion coefficient should adapt to the environment.
This was the main innovation of DFT-D3 compared to D2.

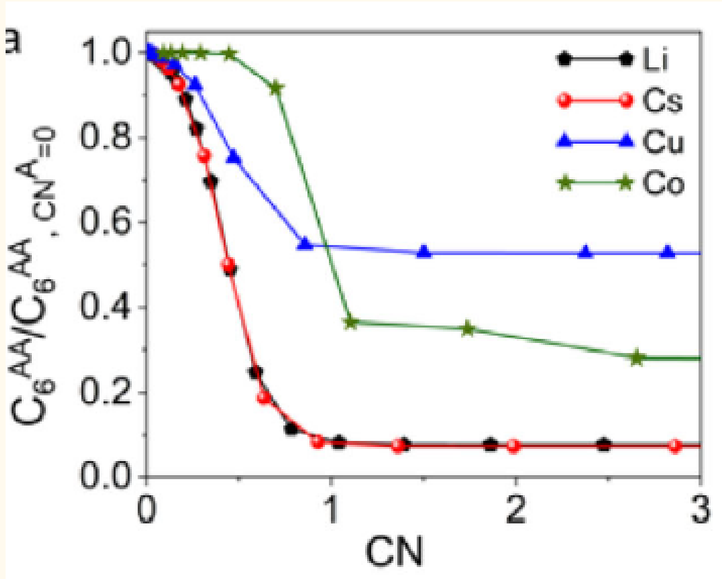
C6 Coefficients:

In D3 the coordination number of an atom is computed using a smooth distance-dependent function and we have seen how to calculate them in the earlier slides.

Coordination number gives us how many neighbors surround an atom

Once the coordination number is known, D3 calculates the dispersion coefficient

Authors plotted dispersion coefficients vs coordination number.

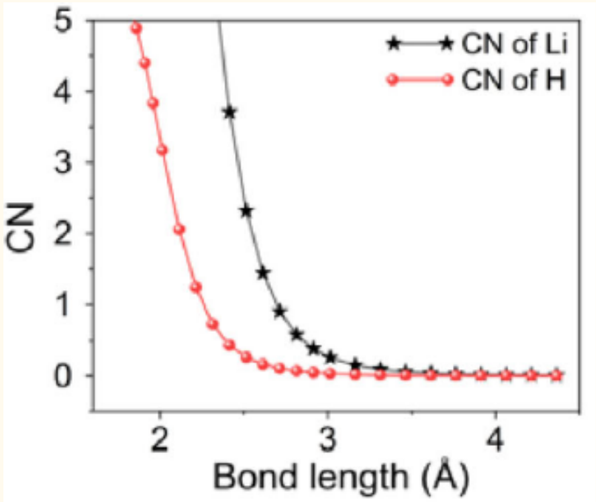


- For highly polarizable metals like Li and Cs,
- The C6 coefficient changes very sharply with CN.

This means small changes in coordination number can cause large changes in dispersion energy.

C6 Coefficients:

The authors also analyzed how CN changes when bond distances vary.



They found that the CN can change very abruptly over a small distance range. for eg in LiH lattice calculations, the CN jumps rapidly around 3.0–3.5 Å

Chain of Events Leading to the Artifact:

Putting these two observations together explains the problem.

Step-by-step mechanism

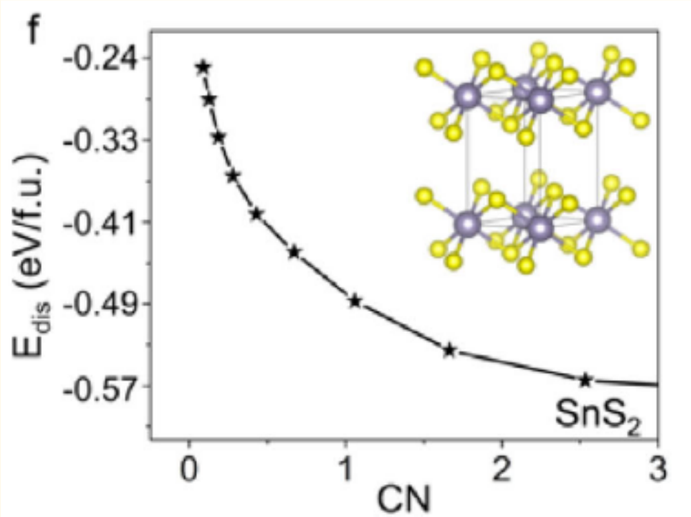
Bond distance changes slightly
↓
Coordination number changes rapidly
↓
C6 dispersion coefficient changes abruptly
↓
Dispersion energy drops suddenly
↓
Artificial energy well appears
↓
Spurious minimum in potential energy surface

Why Metals Are Most Affected and others are not?

- The artifact is strongest for highly polarizable atoms, such as: Lithium, Cesium

:

Highly polarizable atoms have large dispersion coefficients.Therefore any abrupt change in C6 produces large changes in energy. This is why metallic systems show the strongest artifacts.



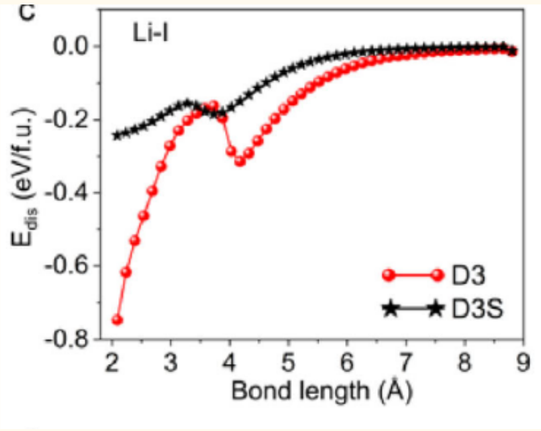
In SnS_2 system the dispersion energy varies smoothly with coordination number. The potential energy surface therefore shows no artificial minimum.

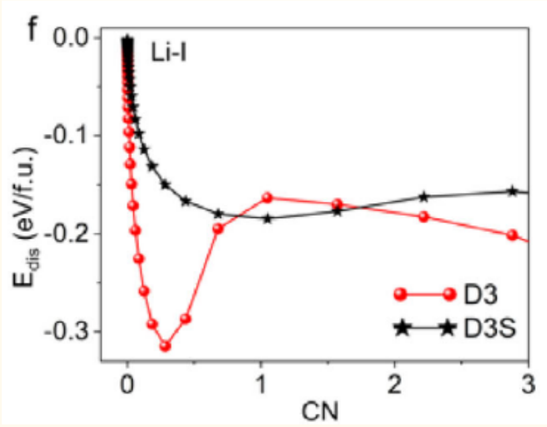
Why D3S Was Proposed:

Because of this issue, researchers proposed a modified model called D3S.

D3S attempts to:

- smooth the interpolation of dispersion coefficients
- reduce abrupt changes in C6.





However, it shows that D3S still produces artifacts in some solid systems.

Why D3S Was Proposed:

The goal of the D3S model is to smooth the dependence of dispersion coefficients on coordination number.

instead of using the same global interpolation parameter for all atom pairs, the D3S approach introduces:

- element-specific interpolation parameters
- pair-dependent smoothing factors

This makes the variation of C6 coefficients with coordination number less abrupt and more physically realistic.

Although D3S was designed to fix the problem, the authors show that it does not completely eliminate the artifact in solids.
Their calculations reveal that:

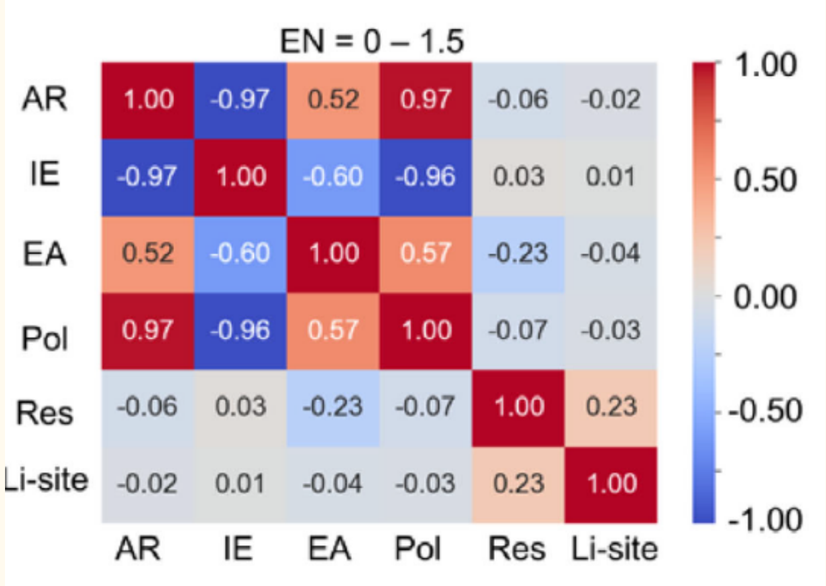
- D3S reduces the magnitude of dispersion errors.
- However, secondary minima can still appear in certain ionic and metallic systems.

Descriptor Correlation Analysis:

Can we predict when D3 will fail using simple chemical descriptors?

Correlations between dispersion errors and properties are examined such as:

- atomic radius
- polarizability
- ionization energy
- electron affinity
- electronegativity
- number of Li atoms in the structure.



it was found that atomic radius and polarizability are strongly correlated in some electronegativity ranges. However no single descriptor reliably predicts the D3 error

Structural and Bonding Motif Effects:

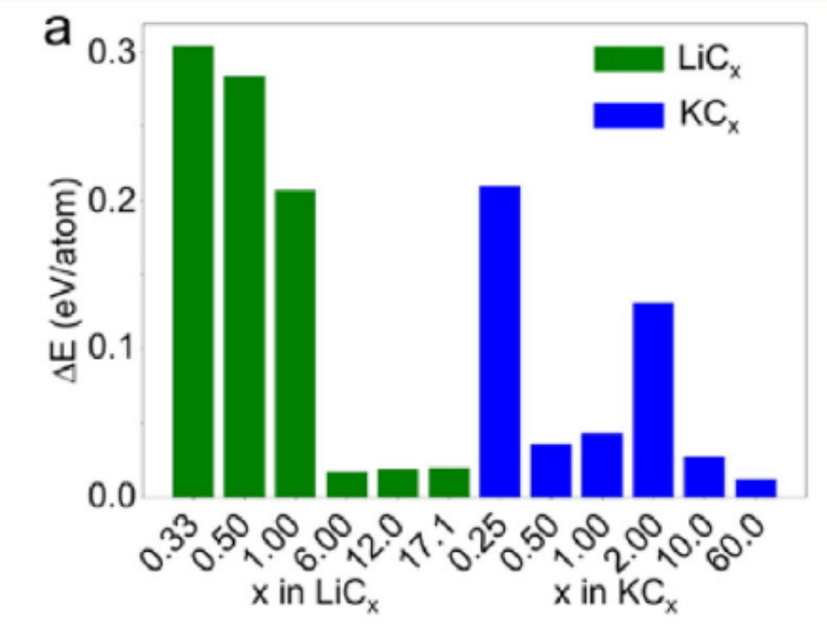
How different crystal structures affect the D3 error.?

Systems such as: LiCx, KCx, LiNx were analysed

It was found that dispersion errors depend strongly on:

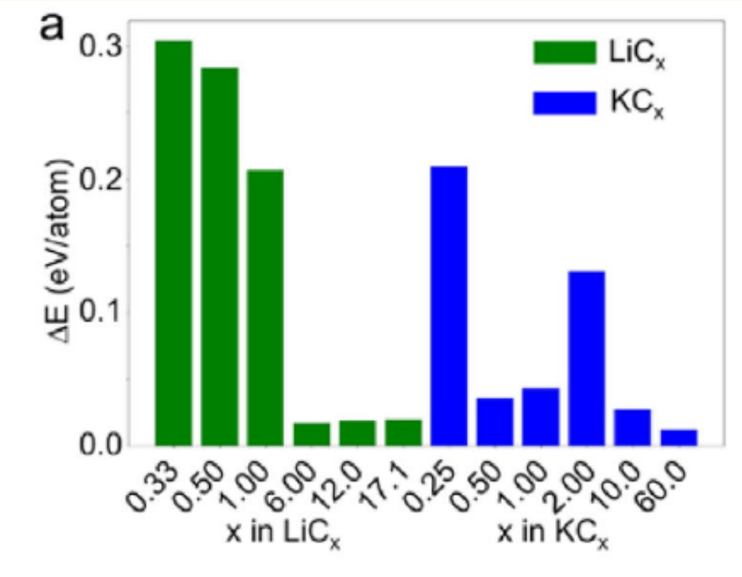
- atomic arrangement
- packing density
- coordination environment

Even compounds with the same chemical formula can have different errors if their structures differ.



Denser structures amplify dispersion overbinding.

Metal Content Effects:



Another key trend the authors observed was that the D3 errors increase with metal content and decrease when the carbon content increases.

Because stronger covalent bonding networks reduce artificial dispersion effects.

Overall Interpretation:

Combining all the results, the authors conclude that the reliability of D3 depends on several factors:

Systems where D3 works well:

- covalent frameworks
- layered materials
- low metal-content structures.

Systems where D3 may fail:

- metallic solids
- ionic compounds
- dense structures with highly polarizable atoms.

Key Findings of the Study:

- The DFT-D3 dispersion correction, although widely used, can produce spurious secondary minima in potential energy surfaces of solids.
- These artifacts arise from the strong dependence of dispersion coefficients C6 on coordination numbers, which may change abruptly with atomic distances.
- The improved D3S model smooths this dependence and reduces the magnitude of the error, but it does not completely eliminate artificial stabilization in certain systems.
- The problem is particularly pronounced in metal-rich, densely packed, or highly polarizable solids.

Conclusion:

The key message of the paper is **DFT-D3** is not universally reliable for solid-state systems.

Although it is computationally efficient and widely used, it can introduce unphysical stabilization in certain materials.

Researchers should therefore use D3 with caution, especially for:

- metallic systems
- dense ionic solids.